

Anthony Okolie

431-278-1445 | Anthonyokolie.ace@gmail.com | [LinkedIn](#) | [GitHub](#) | [my-portfolio](#)

SUMMARY

Software developer with experience in Java, Python, and TypeScript, building scalable, API-driven applications using Spring Boot, Node.js, and PostgreSQL. Solid understanding of distributed systems, microservices architectures, and ACID-compliant database design, with hands-on expertise in Docker, Azure cloud, and CI/CD pipelines. Collaborative team member skilled at translating technical findings into clear insights and driving solutions in financial services environments.

EDUCATION

University of Manitoba

Sep 2021 - Jun 2026

Bachelor of Computer Science, Computer Science

- **Achievements:** Honours
- **Coursework:** Data Structures & Algorithms, Operating Systems, Database Systems, Software Engineering, OOP

SKILLS

- **Languages:** TypeScript, JavaScript, Java, Python, C, C++, Swift, C#, CSS3, Go, Rust
- **Frontend & Templating:** React, HTML5, CSS3, Liquid, templating engines, Vite, Angular
- **Backend:** Node.js, Express, REST APIs, WebSockets, Socket.io, Spring Boot, JWT Authentication, OAuth2.0, IAM concepts, API Design
- **Cloud, Tools & Frameworks:** Docker, CI/CD, Azure, **Linux**, Git, GitHub, VS Code, Kafka, Tableau, AWS, Redux, Npm
- **Databases:** PostgreSQL, SQL, MongoDB, Drizzle ORM
- **Testing:** JUnit, Vitest, Cypress
- **Concepts & Domain Expertise:** Distributed systems, concurrency (**threading and processes**), data consistency, optimistic updates, ACID design, Real-time Solutions, Analytical Abilities, Object-oriented Design Principles, Debugging, Software Development Lifecycle (SDLC), Web applications, **TLS, TCP**

PROJECTS

Real-Time Collaboration Board – Node.js, Express, React.js, Github

- Built a real-time collaboration platform in TypeScript with Node.js and Express, supporting concurrent multi-user interactions in a distributed environment
- Implemented optimistic UI updates with WebSocket channels to enable real-time solutions and ensure low-latency synchronization across concurrent users
- Designed and developed REST APIs with Express.js for consistent state management, emphasizing API best practices and data consistency
- Integrated JWT-based authentication and reusable UI templates to streamline component rendering.
- Configured a CI/CD pipeline with Docker and AWS Elastic Beanstalk for automated deployments, and hosted Postgres on Amazon RDS
- Developed Tableau dashboards during academic assignments to visualize trends, perform exploratory data analysis, and communicate insights to non-technical audiences.

Point-of-Sale (POS) System – TamboAI, SpringBoot, Docker, Github

- Developed microservices-based backend services using Spring Boot, containerized with Docker to handle inventory, carts, sales, and customer management
- Enforced business rules using PostgreSQL constraints and triggers, ensuring ACID compliance and data integrity
- Built type-safe REST APIs with Zod validation; integrated Generative UI via Tambo AI.
- Used templated UI components and dynamic rendering patterns similar to Liquid-style workflows.

Trade Execution & Risk Monitoring Microservice – SpringBoot, Typescript, PostgreSQL, Docker, Github

- Designed a low-latency trade execution microservice with end-to-end workflows including validation, risk checks, and execution under concurrent load.
- Engineered ACID-compliant PostgreSQL schemas and optimized queries, reducing validation latency by ~30%.
- Implemented event-driven architecture with WebSockets and containerized the database using Docker for portability.

Real-Time Collaborative Document Editor – Angular, Node.js, WebSockets, Redis, Github

- Built a scalable real-time collaborative document editor using Angular, Node.js, and WebSockets, enabling concurrent multi-user editing with low-latency synchronization
- Implemented conflict resolution using operational transformation concepts to ensure consistency across distributed clients
- Designed a horizontally scalable backend using Redis Pub/Sub to synchronize state across multiple server instances
- Engineered real-time presence features including live cursors and user session tracking
- Improved system reliability by handling reconnections, out-of-order events, and duplicate message delivery
- Implemented multithreaded and concurrent patterns in distributed real-time systems

EXPERIENCE

DNOJ Accounting and Tax

May 2025 - Jul 2025

Web Designer

- Cleaned, validated, and transformed raw financial data for reporting in a financial services environment, applying analytical abilities and debugging techniques to ensure accuracy.
- Built structured data flows and documentation to support dashboard creation, leveraging version control systems for collaboration.
- Communicated insights and technical findings to non-technical users, demonstrating strong analytical abilities.

Canada Revenue Agency

May 2022 - Dec 2022

T1 Matching Assessor

- Processed 1,000+ tax cases with 90%+ accuracy in a financial services environment, leveraging debugging techniques to identify discrepancies.

- Communicated with taxpayers to request missing documentation, ensuring compliance and timely resolution.